Object-Oriented Programming

Final Project: Space Crew Game for Android

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GitHub
Video Explanation

0 Summary of Features

0.1 Minimum Requirements

- Turn-Based Loop: A strategic layer for managing the crew on a 30-turn mission.
- Crew Management: Recruit from 5 specialized classes, train, heal, dismiss.
- Training System: Assign crew members to training to gain XP and improve their efficiency and max HP.
- Crew Recovery: Slow passive healing in Quarters, faster in Medbay.
- Cooperative Mission System: Crew members take turns acting against the threat, the threat retaliates after the player's actions
- Scaling Difficulty: Threat HP and Attack power scale linearly as the mission progresses.
- Mission Goal: Survive until Turn 30 to complete the mission.
- Text-Based Logic: Monospace typography and ASCII-style formatting for a command-line commander experience.

0.2 Bonus Features

- Limited Actions: Ship actions are limited to 2 per turn, requiring prioritization.
- No Death: Automatic transition to medbay for crew members incapacitated in combat. A triage system automatically bumps the healthiest patient to quarters to make room for critical 0-HP cases.
- Vitality Decay: Every time a crew member is incapacitated (0 HP), they suffer a permanent -3 penalty to their Max HP. Crew members whose Max HP drops to 5 or lower are permanently retired from service.
- Tactical Combat: Each crew member has 3 actions to choose from, with varying effectiveness against threats
- Cooldown Mechanics: Powerful actions have 1–3 round cooldowns to prevent spamming.
- Threat generated at random from 5 different Threat Types: Boarding Parties, Ship Hazards, Debris Fields, Pirate Raiders, and Faction Standoffs.
- Elite Encounters: 15% chance for "Elite" variants with 30% higher stats.
- Selective Targeting: Threats distribute damage between active crew but refocus all power on a single survivor if one is downed.
- Specialized Classes: 5 classes with varying strengths against threat types
- Leveling System: Crew gains XP from combat and training. Leveling up increases skill efficiency, grants a full heal, and a permanent +4 boost to Max HP.
- Scoring system: A final score is calculated based on turns survived, threats neutralized, total crew levels, total damage taken, and permanent casualties.
- Recycler View: renders individual crew members in the Ship view and crew actions in the Combat view
- Retro UI: A consistent terminal retro-gaming visual style with black backgrounds and high-contrast green text.

1 Gameplay

The game is a survival game in which the player must patrol a sector for 30 turns while encountering various threats. The game is over if the crew is incapacitated and can't counter a new threat. Various factors contribute to the player's score: turns survived, threats resolved, levels gained, damage taken, and crew members lost.

1.1 Actions in ship time

There are two levels of time in the game: ship time and combat time. In ship-time, the player manages the crew by recruiting/dismissing, training, and healing them. The player can perform up to 2 actions per turn. Every turn in ship-time ends with patrol duty, during which there is a chance of encountering a threat. Once a threat emerges and the player chooses two crew members to fight it off, the game transitions to combat time. All combat rounds take place within one ship turn.

1.2 Actions in combat time

Each crew member has a set of three actions they can perform during combat. The crew members will perform their actions in the order they were picked for combat duty, which matters for some actions that boost the ally's action. Many actions have a cooldown period, so the player has to vary their tactic, making the combat more interesting. A crew member can also skip the turn by not choosing an action if no good options are available. Once the threat is resolved or both crew members are incapacitated, the game returns to ship time.

1.3 Experience, Leveling, and Skills

Crew members gain experience through combat (+25xp) or training (+50). Once crew members have accumulated enough XP (LV*100), they level up. Level is a multiplier for three main areas: success chance, damage/healing, and damage reduction. For certain skills, such as a diplomat's 'Negotiate' and an engineer's 'Isolate Systems', skill level increases the chance of success. For example, a level 1 diplomat has a 35% chance to negotiate a cease-fire, which peaks at 85% at level 6. Each new level also increases the max HP of a crew member.

1.4 Healing

Crew members heal a small amount of HP each ship turn (unless incapacitated). An injured crew member can also be assigned to medbay for faster healing. Crew members who become incapacitated in combat will automatically be assigned to medbay at the end of the turn. Since the medbay can only hold 2 crew members at a time, a medical triage system ensures that the patients with the most HP make room for incapacitated crew members. However, incapacitation also permanently lowers crew members' max HP, causing their vitality to decay over time. Crew members with 5 or fewer max HP aren't fit for combat and will be retired.

2 Threats

The games have five types of threats that are equally strong in HP and ATK, but the best counter tactics vary.

Threat	Description	Best counter tactics
Debris Field	Immovable force, evade or blast them.	High-G maneuver (Pilot), Weapon Systems (Marine)
Ship Hazard	Internal tech threat, evasion and guns are useless.	Isolate Systems (Engineer), Close Combat (Marine)
Pirate Raider	Fights dirty and respects force.	Weapon Systems (Marine), Ramming Approach (Pilot)
Boarding Party	Perimeter breached, this could get ugly.	High-G Maneuver (Pilot), Close Combat (Marine)
Faction Encounter	Peaceful solution is an option.	Negotiate (Diplomat), Ramming Approach (Pilot)

2.1 Threat Scaling

As the game progresses, the threats get more challenging. The core stats are calculated based on the current Ship Turn:

- Base HP: 12 + (Turn * 2)
- Base Attack: 4 + Turn

Any threat can spawn as a 30% stronger Elite variant (15% chance):

- Elite HP: Base HP * 1.3
- Elite Attack: Base Attack * 1.3

2.2 Damage distribution

Threats distribute their damage equally to all standing crew members. As soon as one of them is incapacitated, all damage will be dealt to the remaining crew member.

3 Classes

There are five crew member classes: Pilot, Engineer, Marine, Scientist, and Diplomat. They differ significantly in base stats and actions. As threats become more challenging, it's crucial to choose the right combination of strengths to counter them.

Class	**Base HP**	**Role**	**Strong against**
Pilot	18	Combat maneuvers & evasion	Faction Standoff, Debris Field
Engineer	16	Shielding, hazard control	Ship Hazard, Boarding Party
Marine	22	Ship-to-ship & close combat	Pirate Raider, Boarding Party
Scientist	14	Buff & heal allies	-
Diplomat	14	Diplomacy & buffing	Faction Standoff

3.1 Actions per class

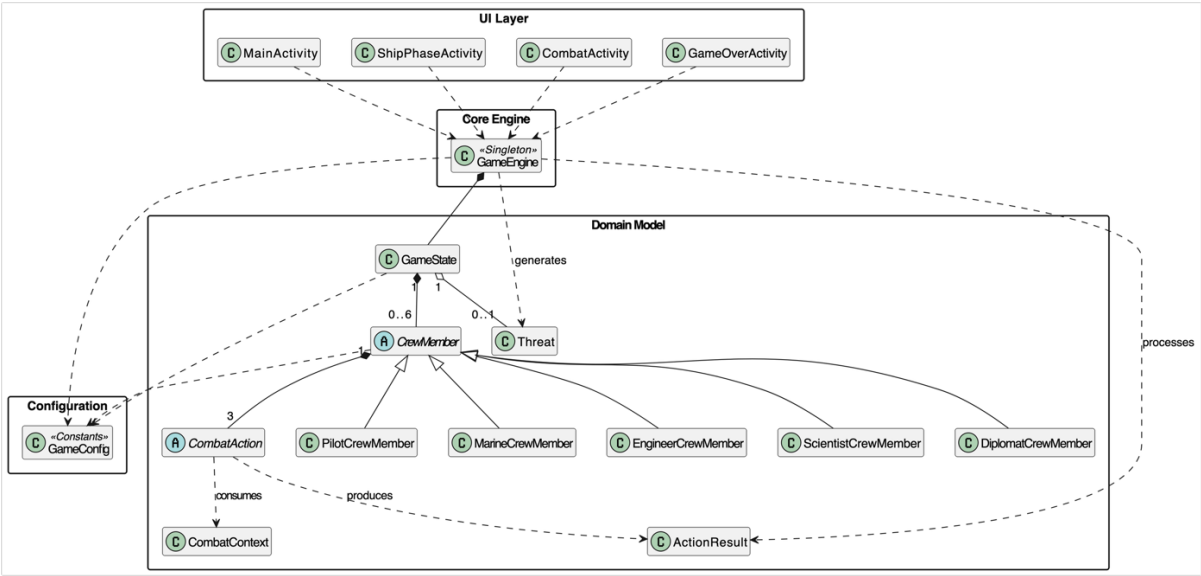
Each class has a unique set of actions, a varying mix of attack (A), defense (D), and utility (U) actions, whose effects depend on the threat type. The table below describes the effects; asterisks indicate relative strength, and an empty cell indicates no effect against that threat.

Pilot	Debris Field	Ship Hazard	Pirate Intercept	Boarding Party	Faction Encounter
A: Ramming approach (cooldown 2)	take damage		**		***
D: Evasive Action	**		**		**
U: High-G Maneuver (cooldown 2)	25…85% to neutralize			25…85% to neutralize	
Engineer	Debris Field	Ship Hazard	Pirate Intercept	Boarding Party	Faction Encounter
A: Overload burst (cooldown 1)	*		*		*
D: Boost shields (cooldown 1)	**		**		**
U: Isolate systems (cooldown 1)		30…90% chance to neutralize		stuns for 1 turn	
Marine	Debris Field	Ship Hazard	Pirate Intercept	Boarding Party	Faction Encounter
A: Weapon systems	**		***		**
D: Fortify (cooldown 2)	*		*	***	*
A: Close combat (cooldown 1)		** take damage		*** take damage	
Diplomat	Debris Field	Ship Hazard	Pirate Intercept	Boarding Party	Faction Encounter
A: Intimidate (cooldown 2)			45…100% to stun		45…100% to stun
U: Negotiate (cooldown 1)			15…75% to neutralize		33…85% to neutralize
U: Tactical Coordination			buff ally	buff ally	buff ally
Scientist	Debris Field	Ship Hazard	Pirate Intercept	Boarding Party	Faction Encounter
A: Hack systems (cooldown 1)			*	*	*
U: Field medicine (cooldown 1)	heal ally	heal ally	heal ally	heal ally	heal ally
U: Analyze (cooldown 1)	buff ally	buff ally			

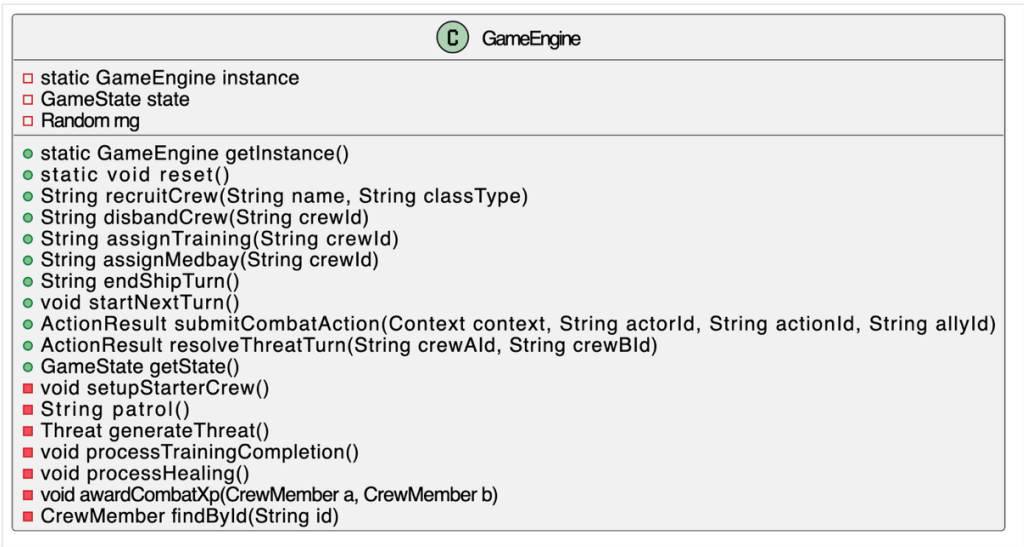
4 UML Diagrams

The game is run by a centralized GameEngine singleton that manages the transition between strategic crew management and tactical combat phases. Relying on the GameState object as the single source of truth, the engine orchestrates crew recruitment, training, and healing. Combat is resolved through a decoupled strategy pattern where specialized CrewMember subclasses execute CombatAction cards against scaling Threat entities. This progression loop rewards survival and tactical efficiency, challenging the player to keep their crew alive across a 30-turn mission for a final calculated command score report.

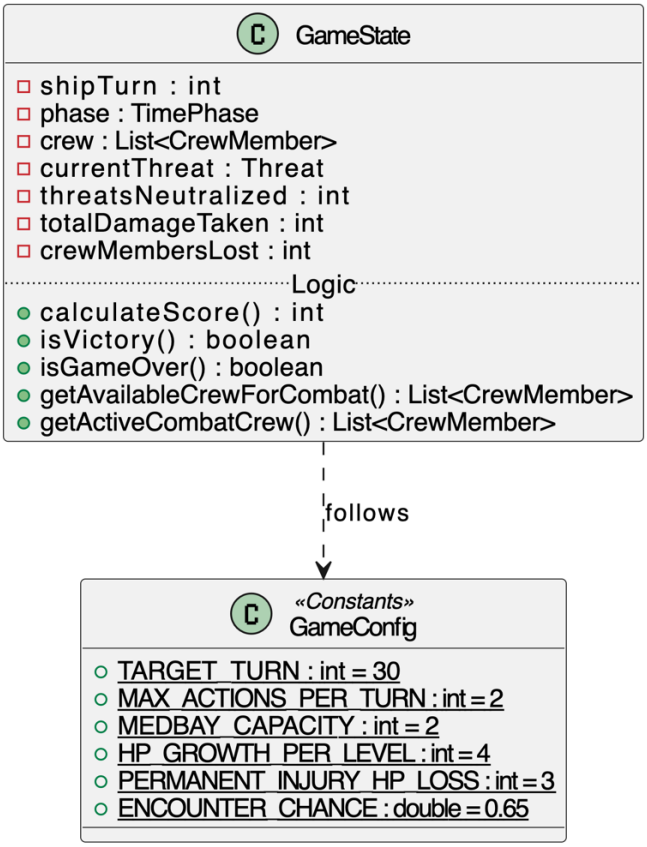
4.1 Overview



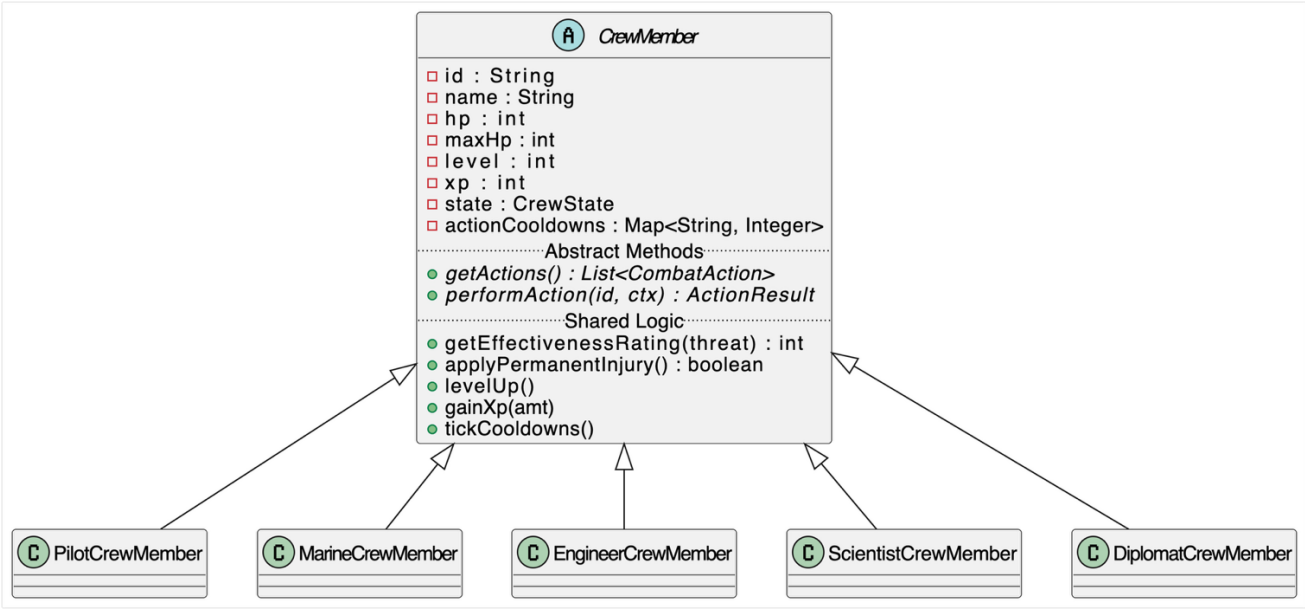
4.2 Game Engine



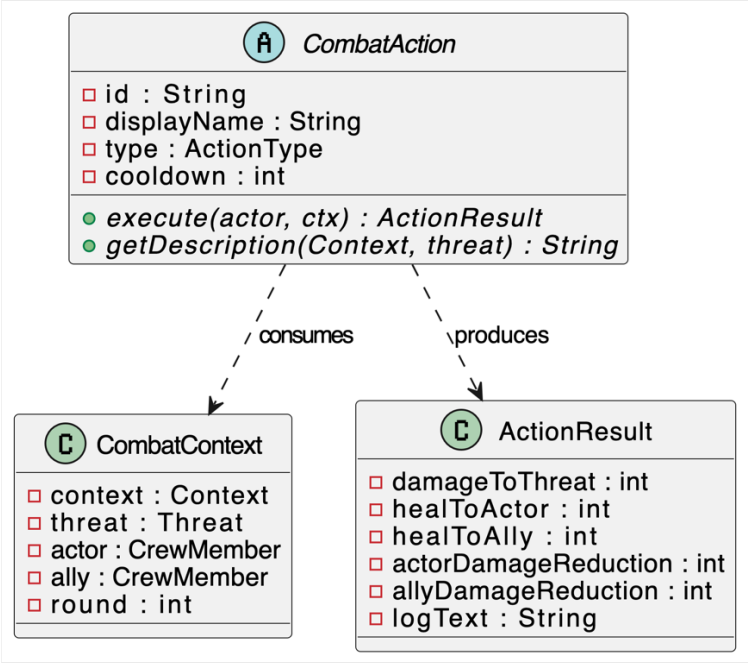
4.3 Game State



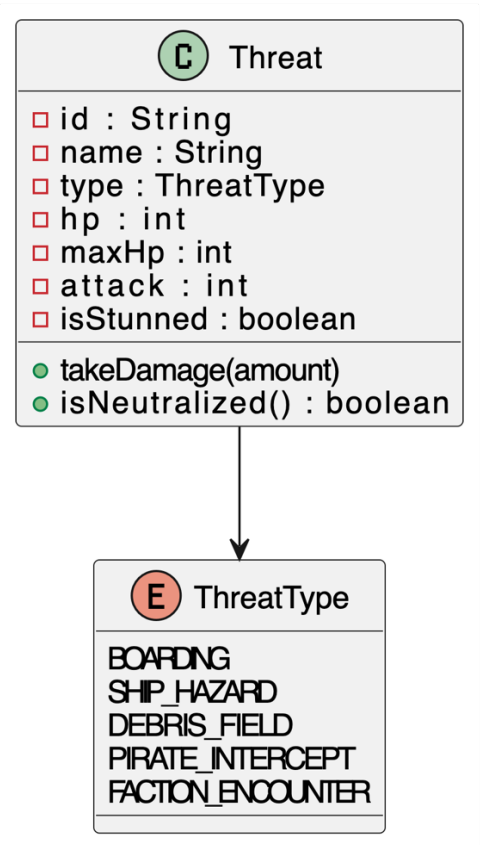
4.4 Crew Member



4.5 Combat Action



4.6 Threat



5 Project Execution and Usage of AI

This has been a solo project.

I have used GitHub Copilot to help design the game architecture and Gemini within Android Studio for advise and help with debugging, refactoring, and documentation.